

PROFICIENT

You have covered all the points.
Keep up with good work.
30/30

Lyla Barr

Homework 2 - BME 100

1. The study proposed in question one brings up several ethical dilemmas. It states that gene therapy may be able to lower levels of cholesterol. This would allow people to consume more fat-rich food with a lower risk of heart disease. This study states that the treatment may not work for everyone, but there is no way of knowing who it would help. Finally, it states that some people would inevitably die prematurely from heart disease. I believe that gene therapy should be approved if the following conditions are true: the gene therapy is targeted to a specific area and will not affect the germline; the group listed in the study that will die prematurely of heart disease (because the gene therapy does not work) are not dying sooner of the disease than they would if the gene therapy was not performed. In other words, the gene therapy would not harm any person who it does not work for, but simply not help them. If these two conditions are true I believe there is no reason why the study should not be approved.

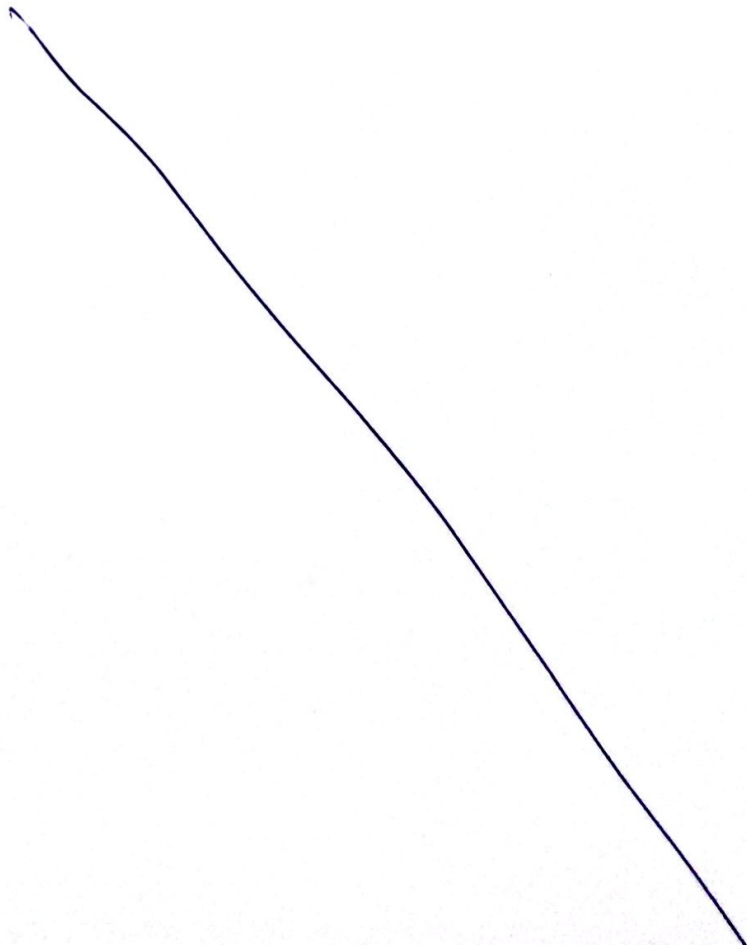
✓ If one of these conditions is false then I believe the study should not be approved. I believe there is no study that would make it acceptable to alter the germline. I also believe there are very few studies that should be approved when they could harm those they do not help. This is not one that should be approved with that condition. Although heart disease is very common and sad, the way the problem is stated in the study suggests that the main reason for the study is to allow people to eat more fat-rich food. Although that would be great if the study did not harm anyone, if the study has a possibility to harm people it should not be approved, when a change in diet is the main reason for the study.

✓ Gene therapy is a difficult ethical subject for several reasons. First, gene therapy is a relatively new technology meaning that research and experience with it is limited. This means that with each study performed with gene therapy, there is an uncertainty factor to what could happen. Secondly, gene therapy can be used in an extremely large range of capacities. It could be used to alter cells such as those in blood or bone marrow. This use can help cure diseases and cannot be carried through a germline. However, gene therapy can be used to make genetic changes that would be passed onto future generations. This application of gene therapy is almost conclusively unethical across the world.

✓ I believe the research community should be very cautious in the use of gene therapy. If used properly, it can have great benefits in curing disease. One large mistake, however, could haunt the community's reputation with the technology, and make patients extremely nervous to partake in it. The community should take each use of gene therapy as a new ethical dilemma. They should weigh the pros and cons of gene therapy with each study, as well as the uncertainty of effects. For instance, gene therapy can not be approved for every study, as this could cause the editing of germlines. Gene therapy should be approved for studies in which it can be isolated to one area and help cure a

disease. Finally, the community should avoid doing any gene therapy that could alter germlines and future generations.

Biomedical engineers should play a large role in shaping positive discussions around the usage of emerging technologies, such as gene therapy. Due to past failures in medical technologies, many people are still extremely nervous of any new technologies in the field. Biomedical engineers should work on rewriting this narrative. However, they must do this in an extremely factual and accurate manner. Within the medical community, biomedical engineers will have some of the greatest knowledge of the technologies. They must list each concern about a technology and ensure that the community stays safe. They must also ensure that they are always giving accurate information based on proven facts.



[Exceeds]

you have covered all the points.
Keep up with good work.

30/30 PM.

Lyla Barr

Homework 2 - BME 100

1. The study proposed in question one brings up several ethical dilemmas. It states that gene therapy may be able to lower levels of cholesterol. This would allow people to consume more fat-rich food with a lower risk of heart disease. This study states that the treatment may not work for everyone, but there is no way of knowing who it would help. Finally, it states that some people would inevitably die prematurely from heart disease. I believe that gene therapy should be approved if the following conditions are true: the gene therapy is targeted to a specific area and will not affect the germline; the group listed in the study that will die prematurely of heart disease (because the gene therapy does not work) are not dying sooner of the disease than they would if the gene therapy was not performed. In other words, the gene therapy would not harm any person who it does not work for, but simply not help them. If these two conditions are true I believe there is no reason why the study should not be approved.

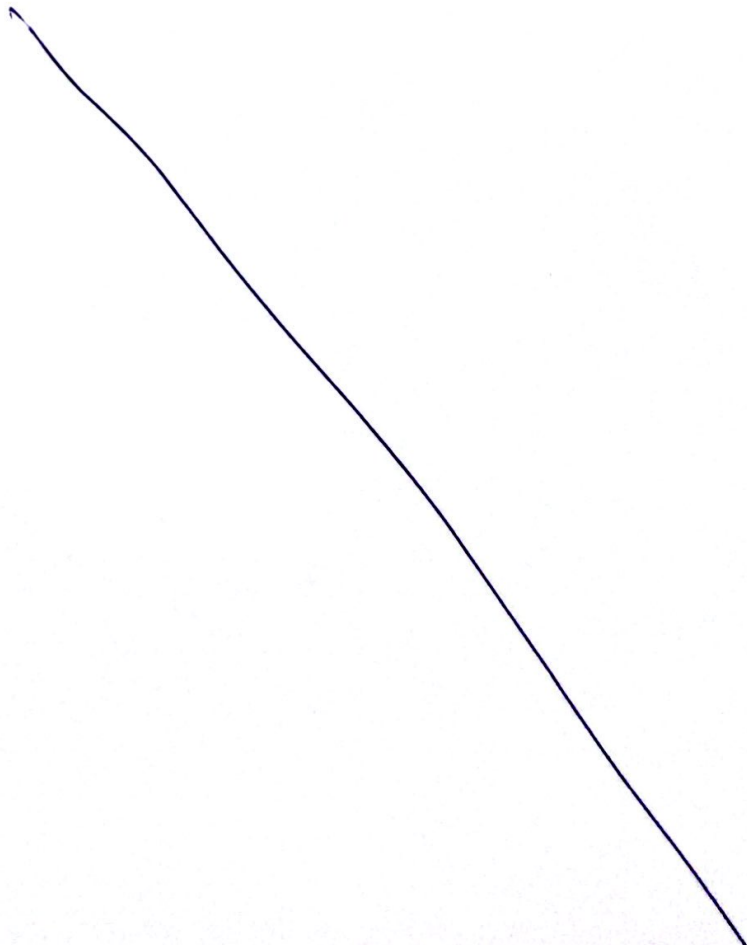
✓ If one of these conditions is false then I believe the study should not be approved. I believe there is no study that would make it acceptable to alter the germline. I also believe there are very few studies that should be approved when they could harm those they do not help. This is not one that should be approved with that condition. Although heart disease is very common and sad, the way the problem is stated in the study suggests that the main reason for the study is to allow people to eat more fat-rich food. Although that would be great if the study did not harm anyone, if the study has a possibility to harm people it should not be approved, when a change in diet is the main reason for the study.

✓ Gene therapy is a difficult ethical subject for several reasons. First, gene therapy is a relatively new technology meaning that research and experience with it is limited. This means that with each study performed with gene therapy, there is an uncertainty factor to what could happen. Secondly, gene therapy can be used in an extremely large range of capacities. It could be used to alter cells such as those in blood or bone marrow. This use can help cure diseases and cannot be carried through a germline. However, gene therapy can be used to make genetic changes that would be passed onto future generations. This application of gene therapy is almost conclusively unethical across the world.

✓ I believe the research community should be very cautious in the use of gene therapy. If used properly, it can have great benefits in curing disease. One large mistake, however, could haunt the community's reputation with the technology, and make patients extremely nervous to partake in it. The community should take each use of gene therapy as a new ethical dilemma. They should weigh the pros and cons of gene therapy with each study, as well as the uncertainty of effects. For instance, gene therapy can not be approved for every study, as this could cause the editing of germlines. Gene therapy should be approved for studies in which it can be isolated to one area and help cure a

disease. Finally, the community should avoid doing any gene therapy that could alter germlines and future generations.

Biomedical engineers should play a large role in shaping positive discussions around the usage of emerging technologies, such as gene therapy. Due to past failures in medical technologies, many people are still extremely nervous of any new technologies in the field. Biomedical engineers should work on rewriting this narrative. However, they must do this in an extremely factual and accurate manner. Within the medical community, biomedical engineers will have some of the greatest knowledge of the technologies. They must list each concern about a technology and ensure that the community stays safe. They must also ensure that they are always giving accurate information based on proven facts.



Homework -BME 100

Please write Q1, on a separate page.

Q1) The first human gene therapy was carried out in 1990 by a group of scientists at the National Institute of Health in Bethesda Maryland. There are many ethical issues and risks associated with gene therapy. Suppose that gene therapy could be used to lower levels of cholesterol and saturated fat in the blood thus allowing people to consume fat-rich fried food with little risk of heart disease. However, this treatment might not work for everyone, and there would be no way of knowing in advance who would benefit. As a result, some people would inevitably die prematurely from heart disease. Should such gene therapy be approved for use in Humans?

No, I believe in this scenario gene therapy should not be approved for use in humans. Permitting the use of gene therapy would create health risks for certain people without their knowledge which is detrimental to society and allowing it to happen is unethical. Currently, people also experience health risks when consuming fat-rich fried food but these are health risks that are known therefore can be avoided with the use of gene therapy. Someone might think that the procedure was effective when it was not and see these health risks when it is too late to fix them. Consuming fat-rich fried foods is also not a necessity which is why I also believe it is not worth risking public health by giving access to gene therapy for the use of something unnecessary.

Your answer is incomplete.

You have not explained how research community and Biomedical engineers should approach

(30 Points) In your mind, what are the major ethical issues raised by gene therapy?

What do you think the research community should do to ensure the ethical use of gene therapy?

What role do you think biomedical engineers should play in shaping discussions around the use of emerging technologies and innovations like gene therapy?

I believe the major ethical issues of gene therapy are:

- Access
- Societal acceptance
- Discrimination
- Distinguishing good and bad uses

I think that the community should continue to talk about gene therapy and look for possible solutions to these ethical problems and even though this technology could help many people we should keep it unattainable to the public until we deal with the ethical issues it may bring. I believe that biomedical engineers should play a very big role in shaping discussions for these emerging technologies as they are the people who understand these technologies best and can understand the effects these technologies have.

Q2) (5 Points) Imagine you have to give a 10-minute presentation to your class on COVID-19 and Diabetes. While preparing the presentation, you find two articles on Diabetes. One article is from Diabetes Res. Clin Pract 2020, and another article is from Diabetes. Education Accredited Program. Which article will you refer to the most and why? Explain with two reasons. (5 Points)

I would choose the article from Diabetes Res. Clin Pract 2020. It's a peer-reviewed journal, meaning it's been checked by experts, and it focuses on clinical research. This means it's more reliable and provides in-depth information on how diabetes affects health, especially during a pandemic.

Q3) (5 Points) If you have to search for a scientific article, why would you search through a library account? Explain with reasons.

I would search through a library account as it provides access to a vast database with many articles, often that are often restricted to users without paying, reliable sources like peer-reviewed journals, and citation management tools that make it easy to save and cite.

Q4) (5 Points) Why do you think sharing a source for the article is important? Give two to three reasons to justify the answer.

It demonstrates that the information you are presenting is based on a reputable source, adding weight to your argument. By providing the source, you allow others to verify the accuracy and reliability of the information themselves. Sharing the source promotes transparency and honesty in your communication.

Q5) (5 Points) We all know MRI is a commonly used imaging modality. What are the other Imaging modalities? List each of them.

- X ray
- CT scan
- PET scan

- Ultrasound

Q6) (5 Points) Imaging modalities help us to understand the tissue or cell. What way does the Imaging modality helps us to understand the tissue or cell?

They provide visual representations of internal structures that cannot be seen with the naked eye. This allows us to observe the size, shape, and arrangement of cells and tissues.

Q7) (5 Points) Last two weeks, you had two guest speakers, Mr. Charles Uth and Professor Kenneth Tichauer. What new information did you learn?

I learned about how to use google scholar and the iit library page. I also learned about the many resources that are available when using the iit library page such as citation websites and online librarians that can guide you through the process of finding information. From Professor Kenneth Tichauer I learned about the many differences between imaging modalities and how each one is better at indicating certain things about the human body.